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Case ID: SOC - 001

Investigating Windows Authentication Activity

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Scenario

The Security Operations Center (SOC) received an alert indicating Windows authentication activity. My task was to investigate whether the failed authentication events represented normal user behavior or suspicious activity by analyzing Windows Security logs in Splunk Enterprise.

What I Did

- I verified that Windows Security logs were available in Splunk.
- I established an authentication baseline.
- I identified accounts with failed authentication attempts.
- I compared successful and failed authentication events.
- I documented my findings and recommendations.



Evidence 1: Windows Security Log Source

Investigation Question:

Am I investigating the correct log source?

SPL:

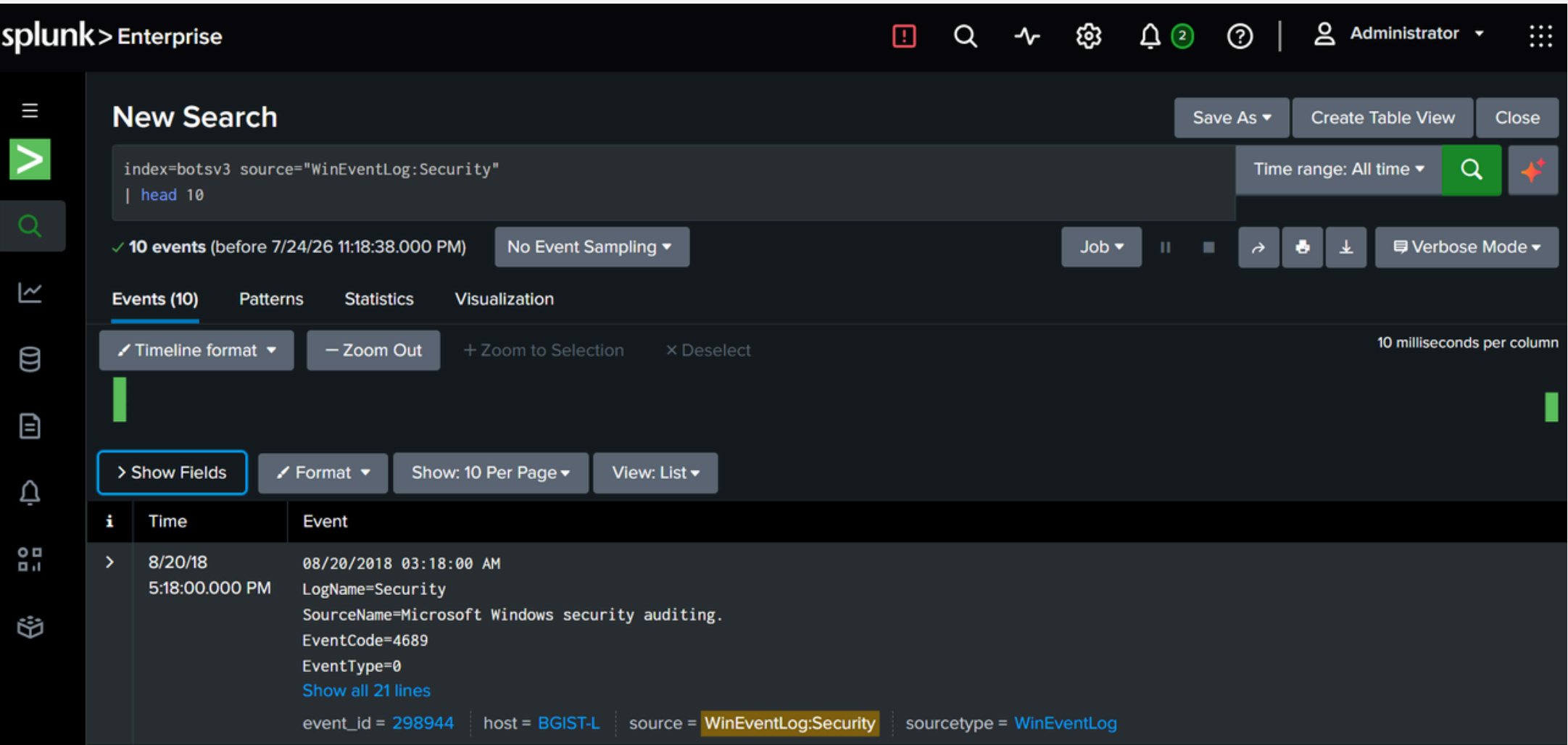
```
index=botsv3 source="WinEventLog:Security"
| head 10
```

What I Found

I confirmed that Windows Security Event Logs were available in the botsv3 index and could be used for the investigation.

Screenshot

Figure 1: Windows Security events available in Splunk.



Evidence 2: Authentication Baseline

Investigation Question:

How many successful and failed authentication events exist?

SPL:

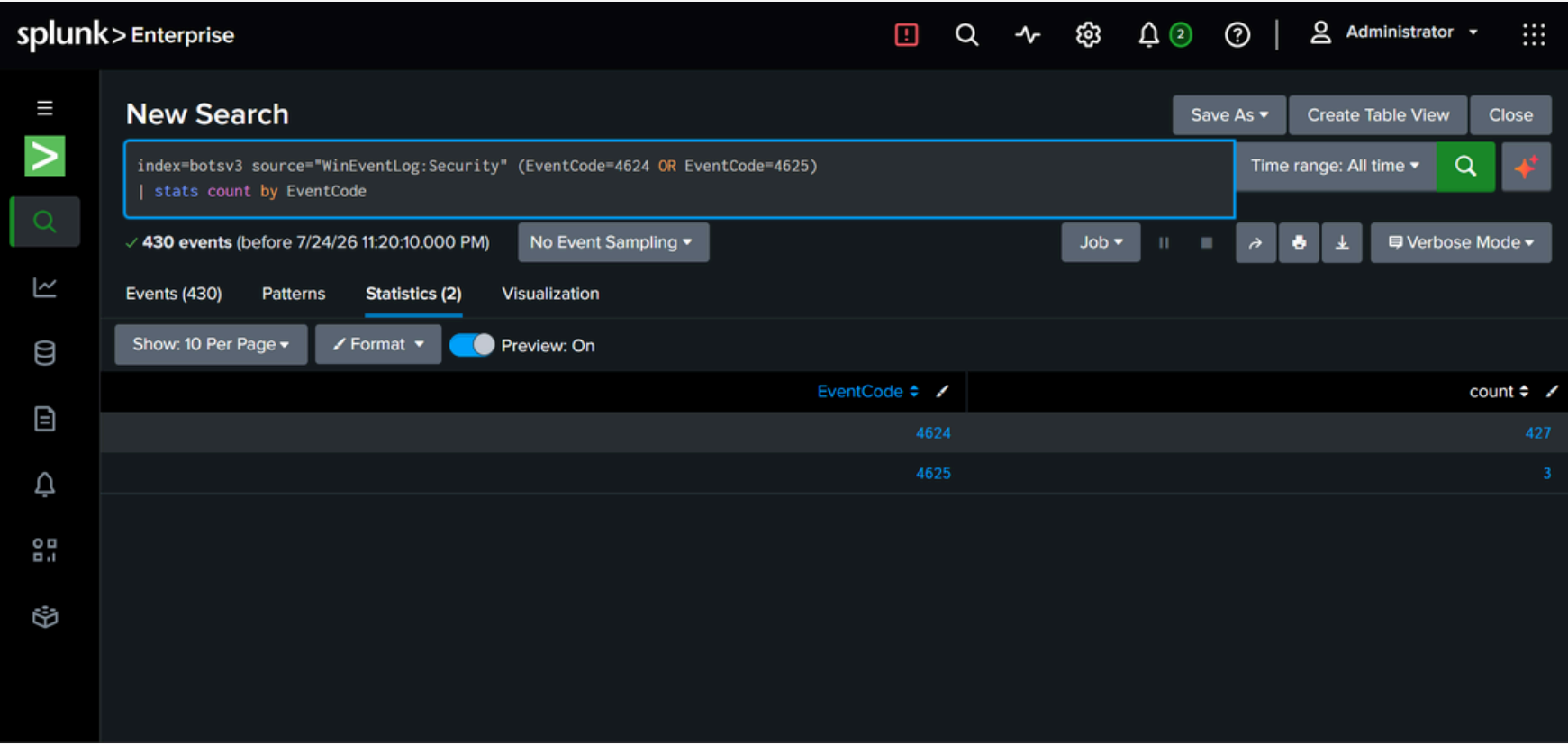
```
index=botsv3 source="WinEventLog:Security"
(EventCode=4624 OR EventCode=4625)
| stats count by EventCode
```

What I Found

I established an authentication baseline by reviewing Windows Event IDs 4624 and 4625. The results showed 427 successful logons and 3 failed logons, providing a starting point for the investigation.

Screenshot

Figure 2 – Authentication baseline.



Evidence 3: Failed Authentication Accounts

Investigation Question:

Which accounts experienced failed authentication?

SPL:

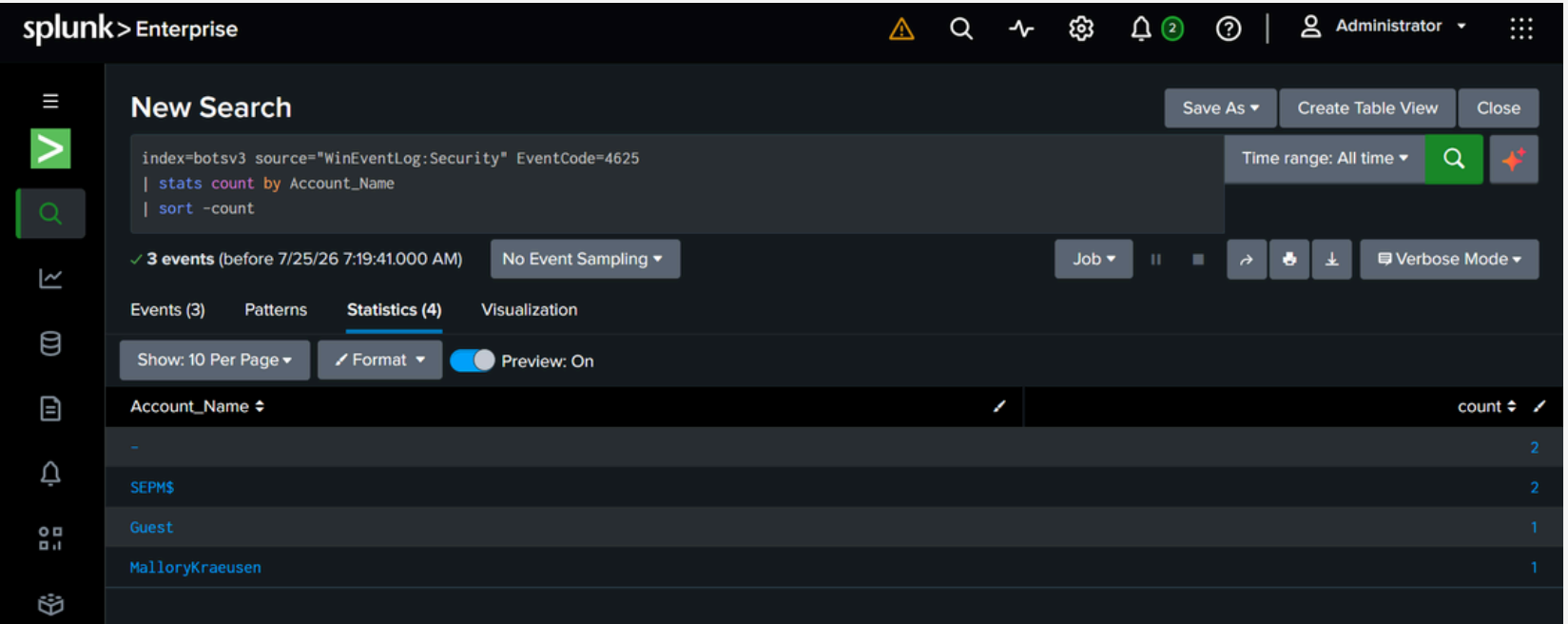
```
index=botsv3 source="WinEventLog:Security" EventCode=4625
| stats count by Account_Name
| sort -count
```

What I Found

I identified the accounts associated with failed authentication events and confirmed that the failures were distributed across multiple accounts.

Screenshot

Figure 3 – Accounts with failed authentication events.



Evidence 4: Authentication Correlation

Investigation Question:

Did any failed authentication attempts result in successful logons for the same account?

SPL:

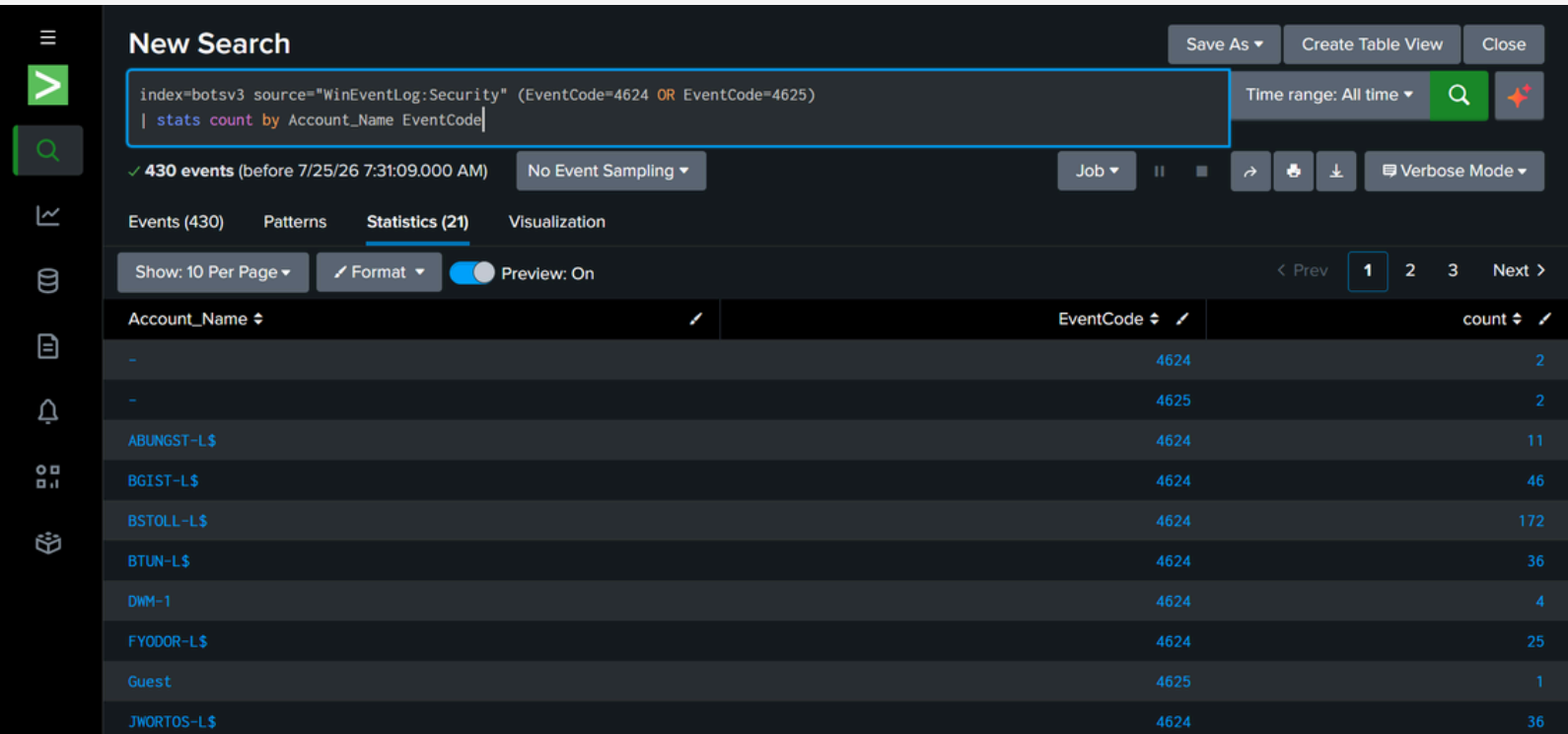
```
index=botsv3 source="WinEventLog:Security" (EventCode=4624
OR EventCode=4625)
| stats count by Account_Name EventCode
```

What I Found

I compared successful and failed authentication events by account. Most authentication activity consisted of successful logons (Event ID 4624). Only a small number of failed logons (Event ID 4625) were observed, and they were spread across different accounts rather than concentrated on a single account. Based on the available evidence, I did not identify a pattern suggesting repeated failed authentication attempts against one account.

Screenshot

Figure 4 – Correlation of successful (4624) and failed (4625) authentication events by account.



Findings

1. I verified that the investigation used Windows Security Event Logs.
2. I established an authentication baseline of 427 successful and 3 failed logons.
3. I identified four accounts involved in failed authentication events.
4. I correlated successful and failed authentication events by account.
5. Based on the available evidence, I found no indication that the activity required immediate escalation.

Recommendations

1. Continue monitoring Windows authentication activity.
2. Investigate repeated failed logons against the same account.
3. Correlate authentication events with timestamps and host information if failures increase.
4. Configure a Splunk alert for excessive failed authentication attempts.

Lessons Learned

1. I learned how to investigate Windows authentication events using Splunk.
2. I learned how to establish an authentication baseline before determining whether activity was suspicious.
3. I learned to base conclusions on evidence instead of assumptions.

Skills Demonstrated

Skill	How I Demonstrated It
Windows Event Log Analysis	I analyzed Windows Security Event IDs 4624 and 4625.
Splunk Investigation	I developed SPL queries to collect authentication evidence.
Event Correlation	I compared successful and failed authentication events by account.
Security Analysis	I interpreted authentication data to determine whether escalation was required.
Incident Documentation	I documented findings, recommendations, and lessons learned.